MATLAB R2016a

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SASexp3a.m

```
1   % Experiment 3: Energy and Power Analysis
2   clc; clear; close all;
3
4   dt = 0.001;
5   t = 0:dt:5;
6
7   xE = exp(-t);
8   xP = sin(2*pi*5*t);
9
10  E_num = trapz(t,abs(xE).^2);
11
12  T0 = 1/5;
13  idx = t <= 2;
14  P_num = (1/(t(find(idx,1,'last'))-t(1))) * ...
15      trapz(t(idx),abs(xP(idx)).^2);
16
17  figure;
18  subplot(2,1,1);
19  plot(t,xE); grid on;
20  title('Energy Signal e^{-t}u(t)');
21  xlabel('Time (s)'); ylabel('Amplitude');
22
23  subplot(2,1,2);
24  plot(t,xP); grid on;
25  title('Power Signal sin(2pi5t)');
```

Workspace

Name	Value
dt	1.0000e-03
E_num	0.5000
idx	*1x5001 logical*

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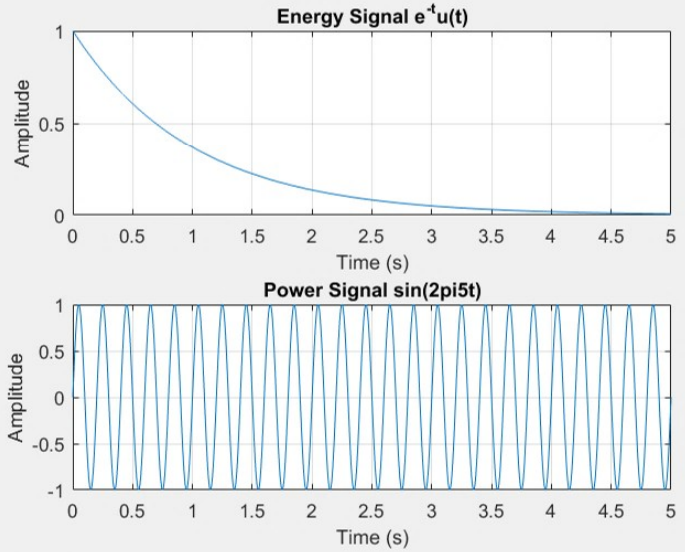
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25  title('Power Signal sin(2pi5t)');
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Figure 1

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Workspace

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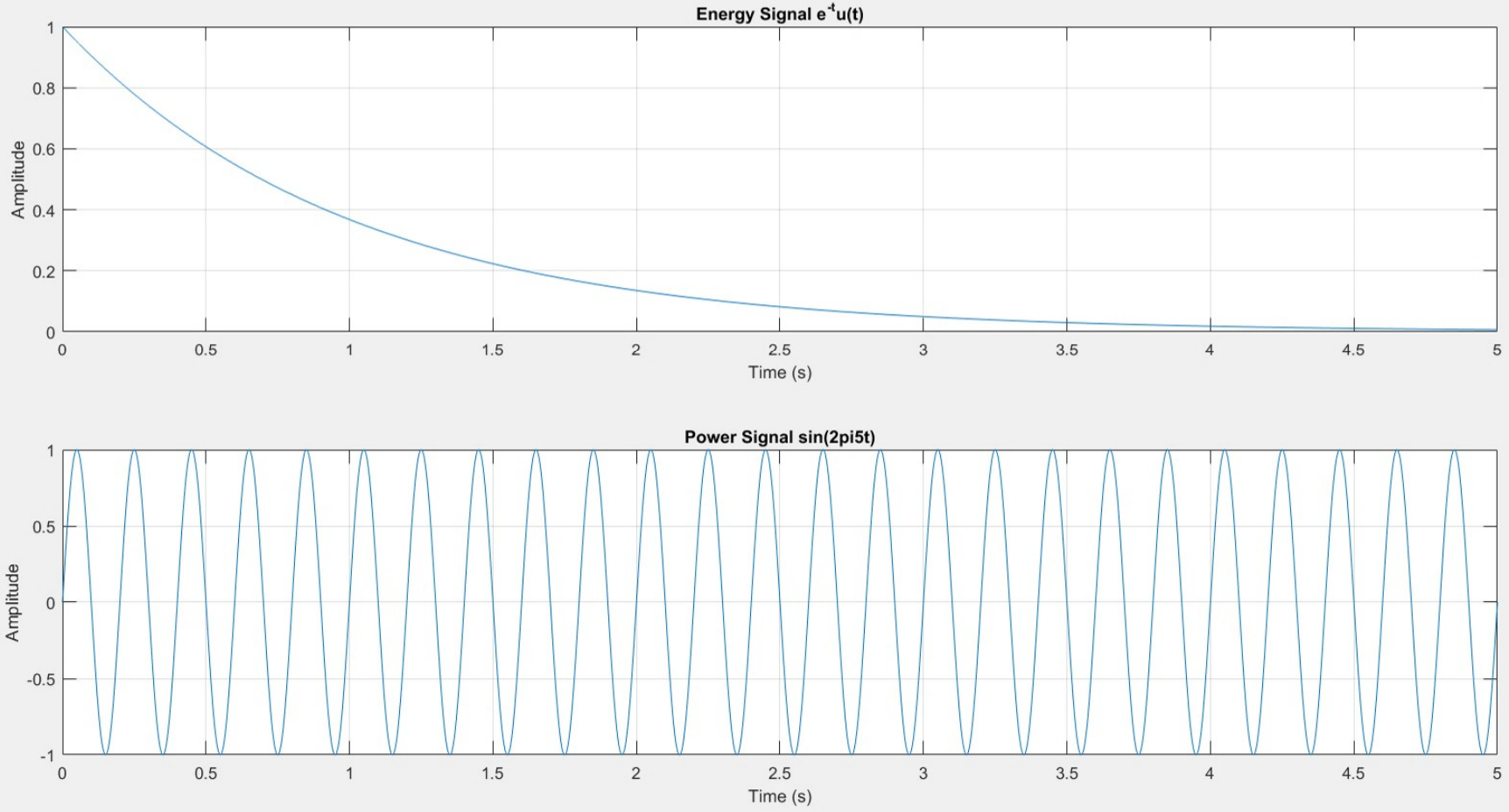
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Figure 1

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